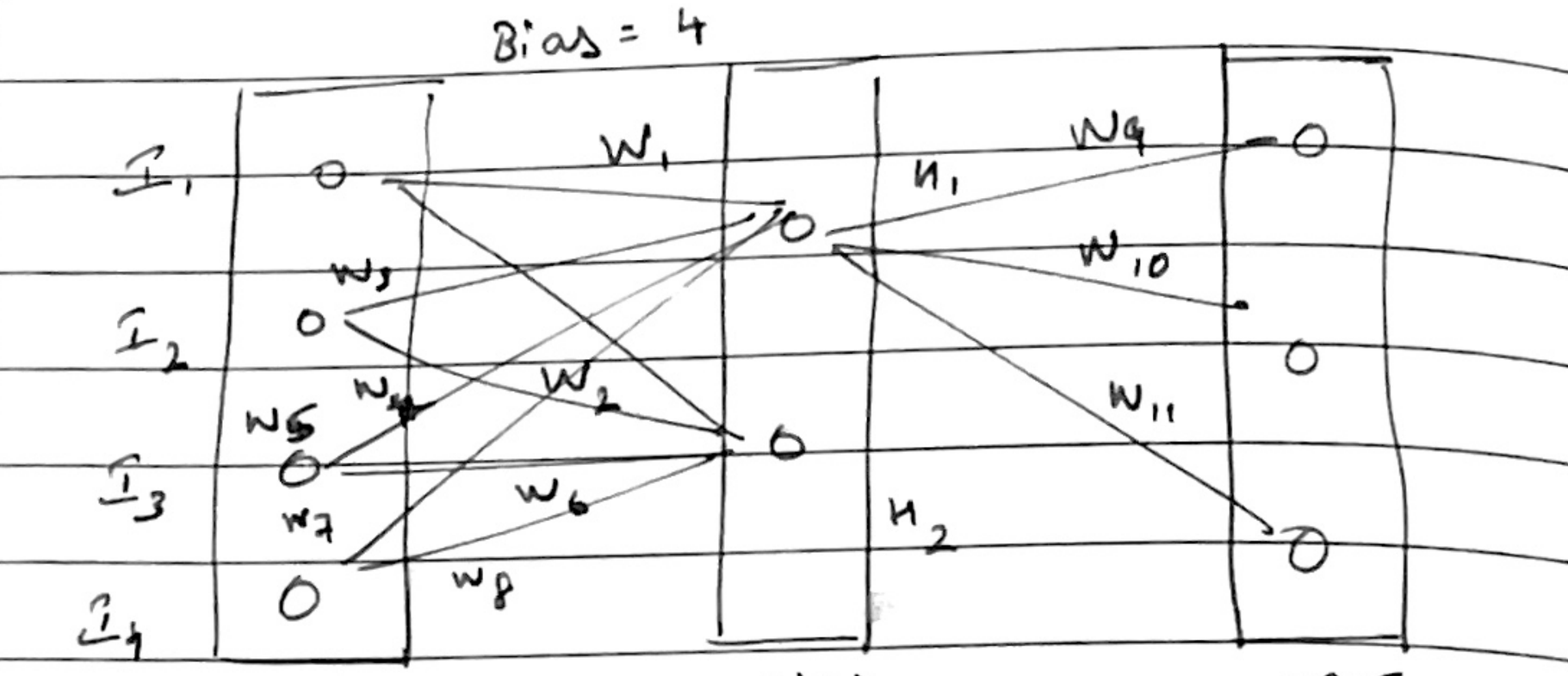
DEEP LEARNING

NEURAL NETWORK ALGORITHM



$$X = [5.1, 3.2, 1.4, 0.2]$$

$$W_1 = 1 \; ; \; W_2 = 2 \; ; \; W_3 = 3 \; ; \; W_4 = 4$$

$$W_5 = 5 \; ; \; W_6 = 6 \; ; \; W_7 = 7 \; ; \; W_8 = 8$$

$$B_1 = 2$$

In the hidden layer :-

$$Z = x_1 W_1 + x_2 W_3 + x_3 W_5 + x_4 W_7 + b$$

$$\Rightarrow (5.1)(1) + (3.2)(3) + (1.4)(5) + (0.2)(7)$$

$$\Rightarrow 5.1 + 9.6 + 7 + 1.4 + 2$$

$$\Rightarrow 12.1 + 11 + 2$$

$$\Rightarrow 23.1 + 2$$

$$\Rightarrow 25.1$$

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★ Activation function :-

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\Rightarrow \sigma(25.1) = \frac{1}{1 + e^{-25.1}}$$

$$\simeq 0.99 \Rightarrow 1 \{\sigma_1\}$$

★ OUTPUT LAYER :-

$w_9 = 1.3$; $w_{10} = 0.5$; $w_{11} = 1.2$

$B_2 = 0.1$

For setosa =>

$$z_{setosa} \Rightarrow \sigma_1 \times w_9 + B_2$$

$$\Rightarrow (1 \times 1.3) + 0.1$$

$$\Rightarrow 1.4$$

For versicolor =>

$$z_{versicolor} \Rightarrow \sigma_1 w_{10} + B_2$$

$$\Rightarrow 1 \times 0.5 + 0.1$$

$$\Rightarrow 0.6$$

For Virginica =>

$$z_{Virginica} \Rightarrow \sigma_1 w_{11} + B_2$$

$$\Rightarrow (1)(1.2) + 0.1$$

$$\Rightarrow 1.3$$

So, $z_{setosa} = 1.4$; $z_{versicolor} = 0.6$; $z_{virginica} = 1.3$